



Title

rdbw2d_dist — Bandwidth selection for distance-based boundary discontinuity designs.

Syntax

```
rdbw2d_dist y distancevar [distancevar ...] [if] [in] [, b(# # [# # ...]) p(#)
kinkunknown(on|off|#) kernel(kernel) bwselect(selector) bwparam(target)
vce(vcetype) bwcheck(#) masspoints(masspointsoption) cluster(clustvar)
scaleregul(#) cqt(#)]
```

Description

rdbw2d_dist computes distance-based bandwidths for **rd2d_dist**. The alias **rdbw2d_distance** is also provided.

Options

b() supplies optional boundary coordinates, one pair per distance variable.

fuzzy() specifies treatment receipt/status for fuzzy bandwidth selection. If supplied, **bwparam()** controls whether the selector targets the fuzzy Wald ratio or the reduced-form outcome discontinuity.

p() sets the local polynomial order. Default is **p(1)**.

kinkunknown() requests the unknown-kink bandwidth-rate adjustment. Use **on**, **off**, or a 0/1 indicator. For bandwidth selection, the first unknown-kink indicator is used.

kernel() specifies the kernel function. Available choices are:
triangular or **tri**: triangular kernel, the default.
epanechnikov or **epa**: Epanechnikov kernel.
uniform or **uni**: uniform kernel.
gaussian or **gau**: Gaussian kernel.

bwselect() specifies the bandwidth selector. Available choices are:
mserd: one common MSE-optimal bandwidth selector for the boundary RD treatment-effect estimator at each evaluation point, the default.
cerird: CER-optimal counterpart of **mserd**.
imserd: IMSE-optimal common bandwidth selector for the boundary RD treatment-effect estimator based on all evaluation points.
icerrd: CER-optimal counterpart of **imserd**.
msetwo: two different MSE-optimal bandwidth selectors, one for negative signed distances and one for nonnegative signed distances, at each evaluation point.
certwo: CER-optimal counterpart of **msetwo**.
imsetwo: two IMSE-optimal bandwidth selectors, one for each side, based on all evaluation points.
icertwo: CER-optimal counterpart of **imsetwo**.

bwparam() specifies the target parameter for fuzzy bandwidth selection. Available choices are:
main: selects bandwidths for the linearized fuzzy Wald ratio, the default.
itt: selects bandwidths for the reduced-form outcome discontinuity. This option is ignored in sharp designs.

vce() specifies the variance-covariance estimator used in bandwidth calculations. Available choices are **hc0**, **hc1**, **hc2**, and **hc3**. Default is **hc1**.

bwcheck() enlarges preliminary bandwidths, when needed, so that at least the requested number of unique observations are included in each side-specific window. Default is **20 + p + 1**.

masspoints() specifies how repeated signed-distance values are handled. Available choices are:
check: detects mass points and reports unique-observation counts, the default.
adjust: adjusts preliminary bandwidths to ensure a minimum number of unique observations within each side of the cutoff.
off: ignores mass points.

cluster() specifies a cluster ID variable for cluster-robust bandwidth calculations.

scaleregul() specifies the scaling factor for the regularization term in bandwidth selection. Default is **scaleregul(1)**.

cqt() specifies the subsample quantile used for initial bias estimation in automatic bandwidth selection. Default is **cqt(.5)**.

Examples

```
. clear
. set obs 800
. set seed 123
. gen x = runiform()*2 - 1
. gen d = x >= 0
. gen y = 1 + 2*x + 1.5*d + rnormal()*0.5
. rdbw2d_dist y x, b(0 0)
```

Stored results

rdbw2d_dist stores **e(bws)**, **e(N)**, **e(p)**, and option labels in **e()**.